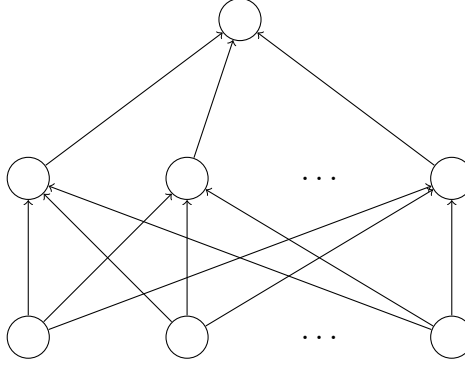


Approximation by Superpositions of a Sigmoidal Function

Notes by Zixin Yu

In 1989, Cybenko published *Approximation by Superpositions of a Sigmoidal Function* [1]. He proved that a feedforward neural network with a single hidden layer and a continuous sigmoidal activation function can approximate any continuous function on the unit cube to arbitrary accuracy. This is a version of the **universal approximation theorem**. In these notes, I outline the proof, clarify points that I found confusing, and expand on details that Cybenko left implicit. The argument uses concepts from measure theory and functional analysis; I introduce the main results as they arise to make the proof easier to follow.

1 Problem Description



A feedforward neural network with a single hidden layer typically takes the following form:

$$\sum_{j=1}^N \alpha_j \sigma(w_j^T x + \theta_j).$$

Here, $x \in \mathbb{R}^n$ is the input, $w_j \in \mathbb{R}^n$ is the input weight vector for hidden neuron j , $\alpha_j \in \mathbb{R}$ is its output weight, and $\theta_j \in \mathbb{R}$ is its bias. The function σ is the activation function, and $x \mapsto w_j^T x + \theta_j$ is an affine function of the input.

Let $I_n = [0, 1]^n$ be the n -dimensional unit cube, and let $C(I_n)$ denote the space of real-valued continuous functions on I_n . Let $M(I_n)$ denote the space of finite signed regular Borel measures on I_n . For $f \in C(I_n)$, the supremum norm is $\|f\|_\infty = \max_{x \in I_n} |f(x)|$. Our goal is to determine when finite sums of the following form are dense in $C(I_n)$ with respect to the supremum norm:

$$G(x) = \sum_{j=1}^N \alpha_j \sigma(w_j^T x + \theta_j).$$

Definition 1

A function $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is *discriminatory* if, for every $\mu \in M(I_n)$, the condition

$$\int_{I_n} \sigma(w^T x + \theta) d\mu(x) = 0$$

for all $w \in \mathbb{R}^n$ and $\theta \in \mathbb{R}$ implies that $\mu = 0$.

I initially found this definition confusing. Equivalently, for every nonzero measure $\mu \in M(I_n)$, there exist $w \in \mathbb{R}^n$ and $\theta \in \mathbb{R}$ such that

$$\int_{I_n} \sigma(w^T x + \theta) d\mu(x) \neq 0.$$

The quantifier “for all w and θ ” is essential: a single vanishing integral does not imply that $\mu = 0$. Intuitively, the family of functions $x \mapsto \sigma(w^T x + \theta)$ can detect every nonzero finite signed regular Borel measure through integration. Now consider the definition of a sigmoidal function.

Definition 2

A sigmoidal activation function satisfies

$$\sigma(t) \rightarrow \begin{cases} 1, & t \rightarrow +\infty \\ 0, & t \rightarrow -\infty. \end{cases}$$

The logistic function $\sigma(t) = 1/(1 + e^{-t})$ satisfies Definition 2. We will work with the general class of sigmoidal functions rather than a particular example. Section 2 presents the main results, and Section 3 applies them to neural networks. Cybenko also discusses other activation functions, which are beyond the scope of these notes.

2 Main Results

Theorem 1

Let σ be any continuous discriminatory function. Then finite sums of the form

$$G(x) = \sum_{j=1}^N \alpha_j \sigma(w_j^T x + \theta_j) \quad (1)$$

are dense in $C(I_n)$. In other words, given any $f \in C(I_n)$ and $\varepsilon > 0$, there exists a function G of the above form such that

$$|G(x) - f(x)| < \varepsilon, \quad x \in I_n.$$

Proof. Let $S \subseteq C(I_n)$ be the set of all finite sums of the form (1). Then S is a linear subspace of $C(I_n)$. To prove density, we must show that its closure $\bar{S}$ is all of $C(I_n)$. Suppose, for a contradiction, that $\bar{S} \neq C(I_n)$. Then $\bar{S}$ is a proper closed subspace of $C(I_n)$.

A Consequence of the Hahn–Banach Theorem

Let X be a real normed vector space, and let Y be a proper closed linear subspace of X . There exists a nonzero bounded linear functional $L : X \rightarrow \mathbb{R}$ such that $L(y) = 0$ for every $y \in Y$.

Applying this consequence of the Hahn–Banach theorem with $X = C(I_n)$ and $Y = \bar{S}$ gives a bounded linear functional L on $C(I_n)$ such that

$$L(h) = 0 \quad \text{for every } h \in \bar{S}, \quad L \neq 0.$$

Riesz Representation Theorem

Let K be a compact Hausdorff space. Every bounded linear functional L on the real-valued continuous functions $C(K)$ has a unique representation

$$L(h) = \int_K h(x) d\mu(x), \quad h \in C(K),$$

where μ is a finite signed regular Borel measure on K .

The Riesz representation theorem gives a measure $\mu \in M(I_n)$ with

$$L(h) = \int_{I_n} h(x) d\mu(x), \quad h \in C(I_n).$$

For every $w \in \mathbb{R}^n$ and $\theta \in \mathbb{R}$, the function $x \mapsto \sigma(w^T x + \theta)$ belongs to S . Since L vanishes on S ,

$$\int_{I_n} \sigma(w^T x + \theta) d\mu(x) = 0.$$

Since σ is discriminatory, $\mu = 0$. This contradicts $L \neq 0$, because

$$\mu = 0 \implies L(h) = \int_{I_n} h(x) d\mu(x) = 0, \quad h \in C(I_n).$$

Therefore, S is dense in $C(I_n)$. □

Lemma 1

Any continuous sigmoidal function is discriminatory.

To prove this lemma, we assume that a measure annihilates every function $x \mapsto \sigma(w^T x + \theta)$ and show that the measure is zero. The key is to rescale the inputs to σ and use dominated convergence to obtain information about the measures of half-spaces.

Proof. Suppose $\mu \in M(I_n)$ satisfies

$$\int_{I_n} \sigma(w^T x + \theta) d\mu(x) = 0 \quad \text{for every } w \in \mathbb{R}^n, \theta \in \mathbb{R}.$$

Fix $w \in \mathbb{R}^n$ and $\theta, \varphi \in \mathbb{R}$. By the sigmoidal limits,

$$\sigma(\lambda(w^T x + \theta) + \varphi) \rightarrow \begin{cases} 1, & w^T x + \theta > 0 \quad \text{as } \lambda \rightarrow +\infty, \\ 0, & w^T x + \theta < 0 \quad \text{as } \lambda \rightarrow +\infty, \\ \sigma(\varphi), & w^T x + \theta = 0 \quad \text{for all } \lambda. \end{cases}$$

Thus, as $\lambda \rightarrow +\infty$, the functions $\sigma_\lambda(x) = \sigma(\lambda(w^T x + \theta) + \varphi)$ converge pointwise to

$$\gamma(x) = \begin{cases} 1, & w^T x + \theta > 0, \\ 0, & w^T x + \theta < 0, \\ \sigma(\varphi), & w^T x + \theta = 0. \end{cases}$$

Define the level set $\Pi_{w,\theta} = \{x \in I_n : w^T x + \theta = 0\}$ and the set $H_{w,\theta} = \{x \in I_n : w^T x + \theta > 0\}$. When $w \neq 0$, these are the intersections of I_n with a hyperplane and an open half-space, respectively.

Because σ is continuous and has finite limits at $\pm\infty$, it is bounded on $\mathbb{R}$. Let $M = \sup_{t \in \mathbb{R}} |\sigma(t)| < \infty$. Then $|\sigma_\lambda(x)| \leq M$, and this constant is integrable with respect to the finite

total variation measure $|\mu|$. Moreover, the hypothesis gives $\int_{I_n} \sigma_\lambda d\mu = 0$ for every $\lambda > 0$, since the rescaled weight and bias are λw and $\lambda\theta + \varphi$. Dominated convergence therefore gives

$$\begin{aligned} 0 &= \lim_{\lambda \rightarrow \infty} \int_{I_n} \sigma_\lambda(x) d\mu(x) \\ &= \int_{I_n} \lim_{\lambda \rightarrow \infty} \sigma_\lambda(x) d\mu(x) \\ &= \int_{I_n} \gamma(x) d\mu(x) \\ &= \sigma(\varphi) \mu(\Pi_{w,\theta}) + \mu(H_{w,\theta}). \end{aligned}$$

This identity holds for every φ . Letting $\varphi \rightarrow -\infty$ yields $\mu(H_{w,\theta}) = 0$, and then letting $\varphi \rightarrow +\infty$ yields $\mu(\Pi_{w,\theta}) = 0$. We now show that these identities force μ to be the zero measure.

Fix w . For a bounded Borel measurable function $h : \mathbb{R} \rightarrow \mathbb{R}$, define the linear functional

$$F(h) = \int_{I_n} h(w^T x) d\mu(x).$$

Since

$$\begin{aligned} |F(h)| &= \left| \int_{I_n} h(w^T x) d\mu(x) \right| \\ &\leq \int_{I_n} |h(w^T x)| d|\mu|(x) \\ &\leq \|h\|_\infty |\mu|(I_n), \end{aligned}$$

the functional F is bounded in the supremum norm. Here $\|h\|_\infty = \sup_{t \in \mathbb{R}} |h(t)|$, and $|\mu|(I_n)$ is finite because μ is a finite signed measure.

Let h be the indicator function of $[\theta, +\infty)$, that is,

$$h(t) = \begin{cases} 1, & t \geq \theta, \\ 0, & t < \theta. \end{cases}$$

Using the identities for the measures of the level sets and half-spaces, we obtain

$$\begin{aligned} F(h) &= \int_{I_n} h(w^T x) d\mu(x) \\ &= \mu(\Pi_{w,-\theta}) + \mu(H_{w,-\theta}) \\ &= 0. \end{aligned}$$

Similarly, if h is the indicator function of $(\theta, +\infty)$, then $F(h) = \mu(H_{w, -\theta}) = 0$. Write h_E for the indicator function of an interval E . For $a < b$,

$$\begin{aligned} h_{[a,b]} &= h_{[a,+\infty)} - h_{(b,+\infty)}, \\ h_{(a,b)} &= h_{(a,+\infty)} - h_{[b,+\infty)}. \end{aligned}$$

By linearity, $F(h_{[a,b]}) = F(h_{(a,b)}) = 0$. The same argument applies to half-open intervals and singletons. Thus, for any finite interval step function $r = \sum_{j=1}^N a_j h_{E_j}$,

$$F(r) = \sum_{j=1}^N a_j F(h_{E_j}) = 0.$$

The set $J_w = \{w^T x : x \in I_n\}$ is a compact interval, possibly a single point. Every continuous function h on J_w can be approximated uniformly there by interval step functions r_k . Extending h and r_k by zero outside J_w allows us to apply F , and the bound

$$|F(h) - F(r_k)| \leq |\mu|(I_n) \sup_{t \in J_w} |h(t) - r_k(t)| \longrightarrow 0$$

shows that $F(h) = 0$. In particular, this applies to the restrictions of sine and cosine to J_w . Since w was arbitrary, the Fourier transform of μ , extended by zero outside I_n , satisfies

$$\begin{aligned} \hat{\mu}(w) &= \int_{I_n} e^{-iw^T x} d\mu(x) \\ &= \int_{I_n} \cos(w^T x) d\mu(x) - i \int_{I_n} \sin(w^T x) d\mu(x) \\ &= F(\cos) - iF(\sin) \\ &= 0, \quad w \in \mathbb{R}^n. \end{aligned}$$

By uniqueness of the Fourier transform for finite measures, $\mu = 0$. Therefore, every continuous sigmoidal function is discriminatory. $\square$

3 Application to Artificial Neural Networks

Theorem 2

Let σ be any continuous sigmoidal function. Then finite sums of the form

$$G(x) = \sum_{j=1}^N \alpha_j \sigma(w_j^T x + \theta_j)$$

are dense in $C(I_n)$. In other words, given any $f \in C(I_n)$ and $\varepsilon > 0$, there exists a function G of the above form such that

$$|G(x) - f(x)| < \varepsilon, \quad x \in I_n.$$

Proof. By Lemma 1, every continuous sigmoidal function is discriminatory, so Theorem 1 applies. $\square$

We have proved that a feedforward neural network with a single hidden layer and a continuous sigmoidal activation function can approximate any continuous function on I_n uniformly. We now turn to classification, a fundamental task in machine learning. Let m denote Lebesgue measure on I_n , and let $P_1, \dots, P_k$ be a measurable partition of I_n : the sets are pairwise disjoint and their union is I_n . Define the decision function f by

$$f(x) = j \quad \text{if and only if} \quad x \in P_j.$$

The value $f(x)$ assigns a class label to x . This function need not be continuous, so Theorem 2 does not apply directly. Theorem 3 shows that a network can approximate f outside a set of arbitrarily small measure.

Theorem 3

Let σ be a continuous sigmoidal function, and let f be the decision function associated with a finite measurable partition of I_n . For any $\varepsilon > 0$, there exists a function of the form

$$G(x) = \sum_{j=1}^N \alpha_j \sigma(w_j^T x + \theta_j)$$

and a measurable set $D \subseteq I_n$ such that $m(D) \geq 1 - \varepsilon$ and

$$|G(x) - f(x)| < \varepsilon, \quad x \in D.$$

Proof. We use the following form of Lusin’s theorem.

Lusin’s Theorem

Let $E \subseteq \mathbb{R}^n$ be a measurable set of finite Lebesgue measure, and let $g : E \rightarrow \mathbb{R}$ be measurable. For every $\delta > 0$, there exists a compact set $D \subseteq E$ such that $m(E \setminus D) < \delta$ and the restriction $g|_D$ is continuous.

Applying Lusin’s theorem with $E = I_n$ and $\delta = \varepsilon$ gives a compact set $D \subseteq I_n$ with $m(D) \geq 1 - \varepsilon$ on which f is continuous. By the Tietze extension theorem, the continuous function $f|_D$ extends to a function $h \in C(I_n)$. By Theorem 2, there is a network function G such that $|G(x) - h(x)| < \varepsilon$ for every $x \in I_n$. Hence, for $x \in D$,

$$|G(x) - f(x)| = |G(x) - h(x)| < \varepsilon.$$

□

4 Conclusion

Hornik’s 1991 paper [2] extended uniform approximation results to continuous, bounded, non-constant activation functions, showing that the sigmoidal limits are not essential. A single hidden layer still suffices, provided that sufficiently many hidden units are available. Universal approximation is an existence result: under the stated assumptions, a network can approximate a target function to arbitrary accuracy. It does not, by itself, specify how many neurons are needed or how to train the network to find suitable weights. Understanding this “universality” therefore requires attention to both approximation theory and practical learning methods.

References

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